



UNIVERSITY OF SRI JAYEWARDENEPURA

Faculty of Technology

Department of Information and Communication Technology

IIC2242 – Advanced Data Structures and Algorithms

Assignment 2

Deadline: 2nd December 2025

1. Implement Min Heap using any programming language. You need to implement the following operations:

- insert(key)
- extractMin()
- decreaseKey(index, newValue)
- delete(key)
- heapify()

Submit a short report explaining your implementation with screenshots/output and provide full source code.

2. Provide a critical analysis of modern hash function improvements. Why do next-generation systems like distributed ledgers (blockchains) and large scale indexing engines require more advanced hashing techniques compared to conventional hashing approaches?